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Scaling SDTM Subsetting: A Simple Program-Based Reusable Template Framework for Compliant, Consistent, and Efficient Delivery

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ABSTRACT

In clinical trials, Study Data Tabulation Model (SDTM) datasets are typically processed collectively across all cohorts within a trial to support efficient downstream activities such as data enrichment, business rule application, and ADaM dataset generation. However, regulatory submissions and interim analyses often require datasets restricted to specific indications or biomarker-defined cohorts, including both screened participants and screen failures. Manually subsetting SDTM domains, or applying macros without a consistent and compliance-focused framework, can introduce redundancy, complexities, inconsistencies, and increased validation effort.

This paper presents a simple, program-based reusable template framework to automate SDTM subsetting for targeted cohorts. The approach applies standardized, domain-agnostic logic across SDTM domains (excluding Trial Design domains) while preserving submission-ready attributes. Key design principles include compliance, efficiency, consistency, and ease of implementation through a lightweight programming approach. Implementation examples from oncology studies demonstrate how the framework enables efficient, consistent, and compliant SDTM subsetting across submission scenarios.

INTRODUCTION

Clinical trial designs have evolved considerably in recent years, particularly in oncology, where basket trials and studies with multiple indications, treatment arms, biomarker-defined populations, or expansion cohorts have become increasingly common. These complex study designs enable the evaluation of multiple patient populations within a single protocol, creating efficiencies in clinical development while increasing the complexity of data management and regulatory reporting. As a result, a single study may support numerous analyses, interim reviews, and regulatory submissions throughout its lifecycle.

To support downstream processes such as data enrichment, standardization, business rule implementation, and ADaM dataset generation, SDTM datasets are typically processed and maintained once for the full study population. This approach provides a consistent data foundation and avoids redundant processing activities across cohorts and analyses. However, regulatory submissions and health authority requests frequently require datasets that are restricted to specific indications, treatment groups, or biomarker-defined cohorts. Depending on the submission strategy, these deliverables may also need to include screened participants and screen failures in addition to enrolled subjects.

This creates a challenge in which a single set of SDTM datasets, originally produced for the full study population, must be repeatedly transformed to support multiple submission packages. For studies with numerous cohorts or indications, the same subsetting activity may need to be performed multiple times throughout the course of development, often across all or a substantial number of SDTM domains.

Traditional approaches to SDTM subsetting can introduce several operational and quality challenges. First, repeated subsetting efforts create programming redundancy and reduce efficiency by requiring similar logic to be developed, validated, and maintained across multiple submissions. Second, independently developed subsetting programs increase the risk of inconsistencies in cohort selection logic or domain processing, potentially leading to discrepancies across deliverables. Finally, each subset dataset necessitates additional review and validation efforts, resulting in a significant validation burden and increased resource requirements to achieve submission readiness.

To address these challenges, a standardized and reusable framework is needed to support compliant, consistent, and efficient SDTM subsetting. This paper presents a simple program-based reusable template framework that automates cohort-specific SDTM subsetting using standardized logic applied across SDTM domains. The approach minimizes repetitive programming, reduces inconsistency risks,

and streamlines validation efforts while maintaining submission-ready datasets, providing a scalable solution for studies requiring multiple submissions or cohort-specific deliverables.

SOLUTION

OVERVIEW

To address the challenges associated with repeated SDTM subsetting for multiple submissions, a reusable template framework was developed to automate the creation of cohort-specific SDTM datasets. The framework is designed to provide a consistent and compliant approach that can be applied across studies, indications, and submission scenarios while minimizing programming effort and validation activities.

The solution adopts a template-driven rather than domain-specific approach. Instead of developing separate programs for each SDTM domain and each submission package, a single reusable template applies standardized filtering logic across the SDTM repository. The framework accepts a cohort definition as input, such as an indication, biomarker-defined population, treatment cohort, or integrated-analysis population. Once the cohort is specified, the template automatically identifies eligible subjects across the relevant SDTM domains and creates the corresponding subset datasets. The approach is applicable to all subject-based SDTM domains while excluding Trial Design domains, which are typically unaffected by cohort selection.

The primary input to the framework is a cohort definition dataset containing the subject population required for a particular submission or analysis. The primary output is a complete set of submission-ready SDTM datasets containing only the requested cohort while maintaining domain structures, variable attributes, and metadata consistency.

By separating cohort definition from dataset processing logic, the framework provides a scalable solution that supports multiple submissions from a single SDTM repository while promoting consistency, compliance, and operational efficiency.

TEMPLATE DESIGN

The framework was developed using several programming principles intended to maximize reusability, consistency, and maintainability.

Consistent Cohort Processing

A key design consideration is the distinction between subject-based eligibility and domain-level processing. Rather than implementing independent filtering logic within each domain program, cohort membership is established once using a centralized cohort definition dataset. This dataset serves as the single source of truth for subject inclusion and is subsequently applied to all SDTM domains.

This design offers several advantages:

- Consistent subject selection across domains
- Reduced risk of programming discrepancies
- Simplified maintenance when cohort definitions change
- Improved traceability and auditability

By identifying the eligible study population once and applying it uniformly throughout the process, the framework ensures that all subsetted domains represent the same cohort.

Attribute Retention

Submission-ready datasets must preserve all required metadata and dataset attributes. Accordingly, the framework is designed to retain:

- Original domain structures

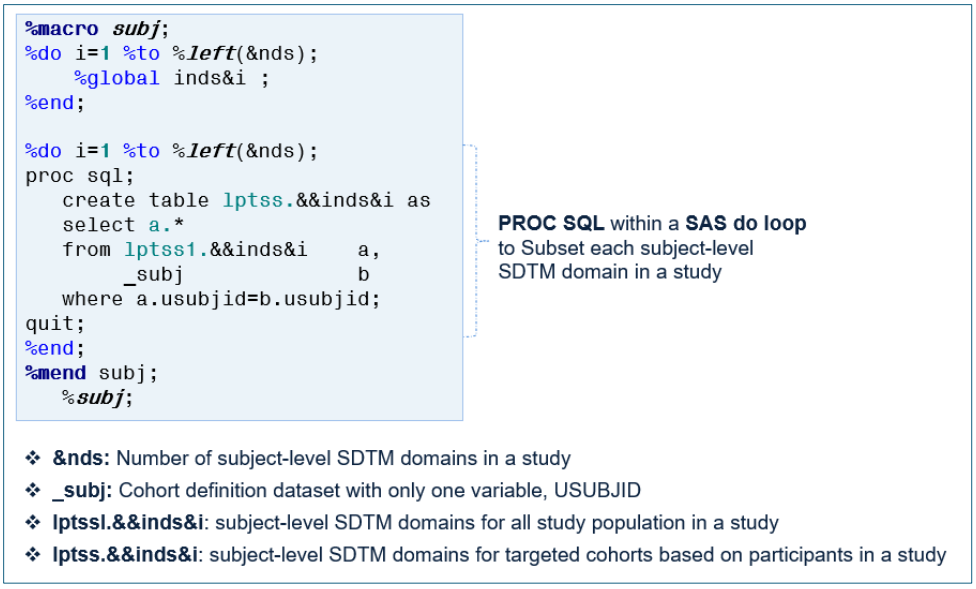
- Dataset-level attributes
- Variable names, labels, lengths, formats and informats

Because the framework subsets existing SDTM datasets rather than recreating them, the resulting output remains aligned with the original submission-standard metadata. This minimizes the risk of introducing structural differences during the subsetting process.

Domain Iteration Using PROC SQL

Because clinical trial data require strict traceability, validation, and adherence to CDISC standards, the choice of data processing approach can affect both efficiency and maintainability. To support consistent SDTM subsetting across multiple domains, the framework employs an iterative processing design using PROC SQL within a SAS macro loop.

Rather than developing separate programs for each SDTM domain, the framework uses a domain inventory to identify the datasets requiring processing and automatically iterates through each domain (Display 1). This approach enables the same subsetting logic to be applied consistently across subject-based SDTM domains while minimizing domain-specific programming.



Display 1 Subject-Based SDTM Domain Subsetting Using PROC SQL and SAS Macro Iteration

PROC SQL was selected because it provides a concise and standardized method for filtering records based on subject membership while preserving existing SDTM dataset structures and attributes. Combined with automated domain iteration, PROC SQL enables a single reusable program to process multiple SDTM domains consistently and efficiently. This domain-agnostic approach minimizes domain-specific programming, reduces long-term maintenance effort, and promotes reusability across multiple submission scenarios.

Here is a comparison of how PROC SQL evaluates against the alternatives.

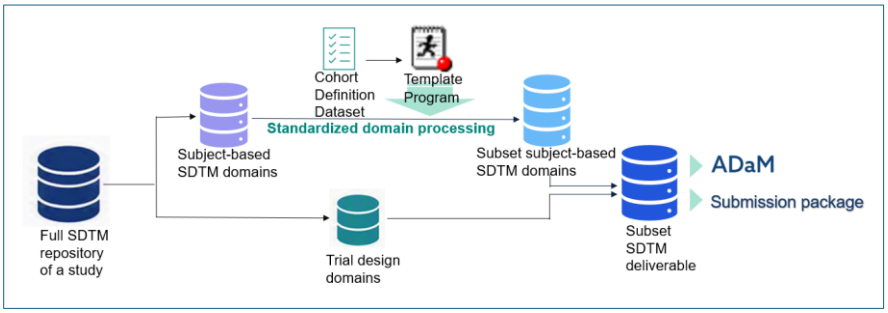
Feature	PROC SQL	SAS DATA Step	R (dplyr)	Python (pandas)
Pre-sorting needed?	No	Yes (for Merges)	No	No
Memory Handling	Disk-based (Excellent)	Disk-based (Excellent)	In-memory (Risk with large data)	In-memory (Risk with large data)
Preserves SAS Attributes?	Automatically	Automatically	Requires conversion	Requires conversion
Code Conciseness	High (for multi-domain joins)	Moderate (requires sort/merge)	High	High

Feature	PROC SQL	SAS DATA Step	R (dplyr)	Python (pandas)
Regulatory Familiarity	Industry Standard	Industry Standard	Growing acceptance	Low/Experimental in formal submissions

Table 1 Key Characteristics of PROC SQL and Alternative Approaches Commonly Used for Data Manipulation

PROCESS FLOW

Display 2 illustrates the overall workflow implemented by the framework.



Display 2 SDTM Domain Subsetting Using the Template Framework

The framework allows the same SDTM repository to support multiple cohort-specific deliverables within a trial through a repeatable and standardized process.

IMPLEMENTATION EXAMPLE

The framework's adaptable design supports implementation across several common submission scenarios, promoting reusability and efficiency.

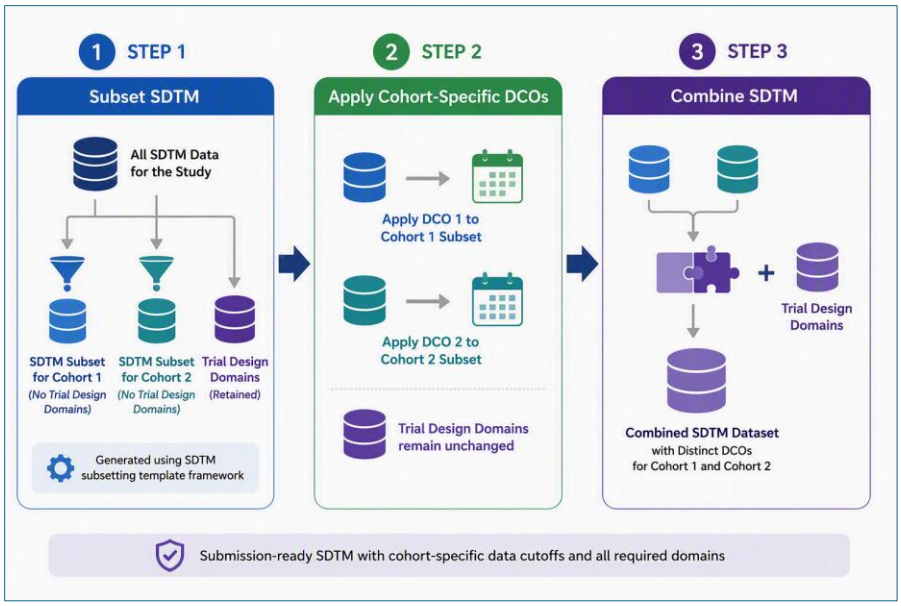
EXAMPLE 1: INDIVIDUAL INDICATION OR BIOMARKER SUBMISSION

In basket studies, individual regulatory submissions may be required for a specific indication or biomarker-defined population. In this scenario, the cohort definition dataset contains subjects belonging to the target indication or biomarker group. The framework automatically subsets all applicable SDTM domains and produces submission-ready SDTM datasets that are included directly in the submission package. This approach eliminates the need for domain-specific subsetting programs and ensures consistent cohort selection across all datasets.

EXAMPLE 2: MULTIPLE BIOMARKER COHORTS WITH DIFFERENT DATA CUTOFFS IN ONE SUBMISSION

Some submission packages for one disease indication may contain multiple biomarker-defined populations with different data cutoff dates (DCOs) within a single deliverable. The framework supports subsetting SDTM data for each biomarker-defined cohort, enabling the appropriate DCO to be applied independently to each cohort. The resulting cohort-specific datasets can then be combined into a single SDTM repository, as illustrated in Display 3.

This approach enables a single execution of the framework to generate a consolidated SDTM package representing the required biomarker-defined cohorts. It maintains consistent filtering logic across domains while preserving dataset structures, metadata attributes, and traceability.



Display 3 Support Cohort-Specific DCO Processing Through the Template Framework

EXAMPLE 3: INTEGRATED SUMMARY POPULATION

Integrated analyses often combine subjects across studies, indications, or development programs to evaluate overall safety. These ADaM datasets commonly focus on treated or safety-population subjects, so screen failures are often excluded. For protocols using SDTM as the source data, the framework can generate SDTM subsets corresponding to the integrated summary population, such as Integrated summary of safety, efficacy and immunogenicity.

These examples demonstrate how a single reusable framework can support diverse submission and analysis requirements while reducing programming redundancy, minimizing inconsistency risks, and lowering validation effort. The result is a scalable and efficient approach for managing SDTM subsetting across complex clinical development programs.

CONCLUSION

Complex clinical trial designs often require multiple cohort-specific regulatory submissions while SDTM datasets are maintained for the full study population. This paper presented a reusable, template-driven framework for SDTM subsetting that applies standardized, domain-agnostic processing logic to generate submission-ready datasets from a single SDTM repository. By centralizing cohort definition and automating domain processing, the framework reduces programming redundancy, improves consistency and traceability, preserves SDTM metadata integrity, and lowers validation effort. The approach is scalable and adaptable to a wide range of submission scenarios, including biomarker-defined populations and multiple data cutoff strategies. Although implemented in SAS, the underlying principles are platform independent and can be applied in other programming environments. Overall, the framework provides an efficient, reproducible, and compliant solution for standardizing SDTM subsetting across increasingly complex clinical development programs.

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